Building a Storage System with Ceph

Ceph answers the need for a free, reliable and distributed storage platform that is scalable to the exabyte level. Ceph's strength lies in its ability to replicate data so that it is fault tolerant. This article takes readers through the steps to setting up a Ceph storage cluster.

Ever since the beginning of mankind, humans have been discovering various means to store their records and data. It all started with the human-rock interaction for storing data. Then we advanced to the human-pen-paper mode of recording data and, as technology improved, the human-computer-tape mode to store data became prevalent. In this article, we will look at how this human-computer-tape mode of data storage is going forward. We have seen that with the increasing development in the field of technology and communication, data sharing between computer systems has increased, and this has led to the building of distributed system storage. We developed the heavy expensive storage appliances like the NetApp FAS3240 and when even these appliances were not big enough for our growing storage needs, we started storing our data on the cloud. The basic foundation of cloud services is built on the computer-network-storage model. Here, we will discuss the latest trend in this area—Ceph, which is open source, freely-available, distributed storage designed to provide excellent performance, reliability and scalability. It uses an object-based storage technique to store and scale data.

Object storage implementation by Ceph

Ceph is a distributed file system built on top of RADOS (reliable autonomic distributed object store). This object store simply stores objects in pools (also referred to as 'buckets'). It is this distributed object store which is the basis of the Ceph file system.

RADOS works with object store daemons (OSD). These OSDs are daemons that work with every storage node in the Ceph cluster. OSD contacts the ceph-mon for cluster membership. This OSD daemon is responsible for read/write operations on the storage disks. This data is stored in the form of objects, and each object corresponds to a file in the file system.

A small RADOS cluster is shown in Figure 1.

Each stored object has a unique identifier, which is unique across the cluster and not only the local file system; metadata, which is a set of key-value pairs; and a single series of bytes of data. The Ceph file system uses the metadata to store file attributes like the owner, dates of creation and modification, etc.

Figure 2 shows Ceph metadata that stores file attributes.

Ceph replaces the centralised gateway with the replication size property. This enables the Ceph OSD to create object replication on each Ceph node. If you choose the number of copies of an object as '5', every object you store on the object pool will be stored on five different OSDs. This provides data safety and availability. Losing one (or more) OSDs will not lead to data loss. The data availability is ensured by the cluster of monitors.

Data placement in RADOS is done by using the CRUSH algorithm. Ceph uses the scalable hashing technique to overcome the older approaches, which enables scale by cleanly distributing work to each OSD daemon on the cluster. With CRUSH, you can strategically place your objects (and its replicas) in different rooms, racks, rows and servers.